Automated cell segmentation with Cellpose-SAM

An approach following Pachitariu, Rariden, Stringer (2025) — *Cellpose-SAM: superhuman generalization for cellular segmentation* (bioRxiv 2025.04.28.651001).

1. Goal

Take a multichannel fluorescence microscopy image — typically DAPI-stained nuclei plus several marker channels — and produce, with minimal user configuration:

- an integer instance mask (one ID per cell),
- a per-cell feature table (morphology + per-channel intensity),
- a visual overlay for quick sanity-checking,
- and a workflow that scales from one image to a folder of thousands.

2. Why Cellpose-SAM

Cellpose-SAM combines the Cellpose flow-field framework with the pretrained ViT-L encoder of Meta's Segment Anything Model (SAM). The result is a foundation model that outperforms inter-annotator agreement on the Cellpose generalist test set, and approaches the human-consensus bound (paper Fig 2d). For users, the practical win is that the model needs almost no per-image tuning:

Property	Paper figure	What it means for the user
Channel-order invariant	Fig 3a	No need to flag which channel is DAPI / cyto.
Size-invariant (7.5–120 px)	Fig 3b	No diameter guessing or size-estimator step.
Robust to Poisson noise	Fig 3c	No denoising preprocessing.
Robust to blur	Fig 3e	No deblurring step.
Robust to downsampling	Fig 3d	No upsampling step.
Robust to anisotropic blur	Fig 3f	No anisotropic-deconvolution step.

These properties together mean the pipeline can be three lines of substantive code: load the image, call `model.eval(...)`, extract features. Anything more is friction.

3. Pipeline

The notebook (`notebooks/cellpose_sam_demo.ipynb`) and the CLI script (`cellpose_sam.py`) implement the same steps:

3.1 Load

Read the image with `tifffile` or `skimage.io`. TIFF, OME-TIFF, PNG, and JPEG all work. 3D volumes are detected from array shape and `do_3D=True` is set automatically.

3.2 Build the model

`models.CellposeModel(gpu=True, pretrained_model="cpsam")` — `cpsam` is the only Cellpose-SAM model. Weights (~370 MB) download once to `~/.cellpose/models/`.

3.3 Segment

Call `model.eval(...)` with paper-default thresholds:

```
masks, flows, styles = model.eval(
    img,
    batch_size=8,
    flow_threshold=0.4,
    cellprob_threshold=0.0,
    normalize=True,
    augment=False,
    tile_overlap=0.1,
)
```

These match the values used in the paper's Methods section for the headline benchmarks.

3.4 Quantify

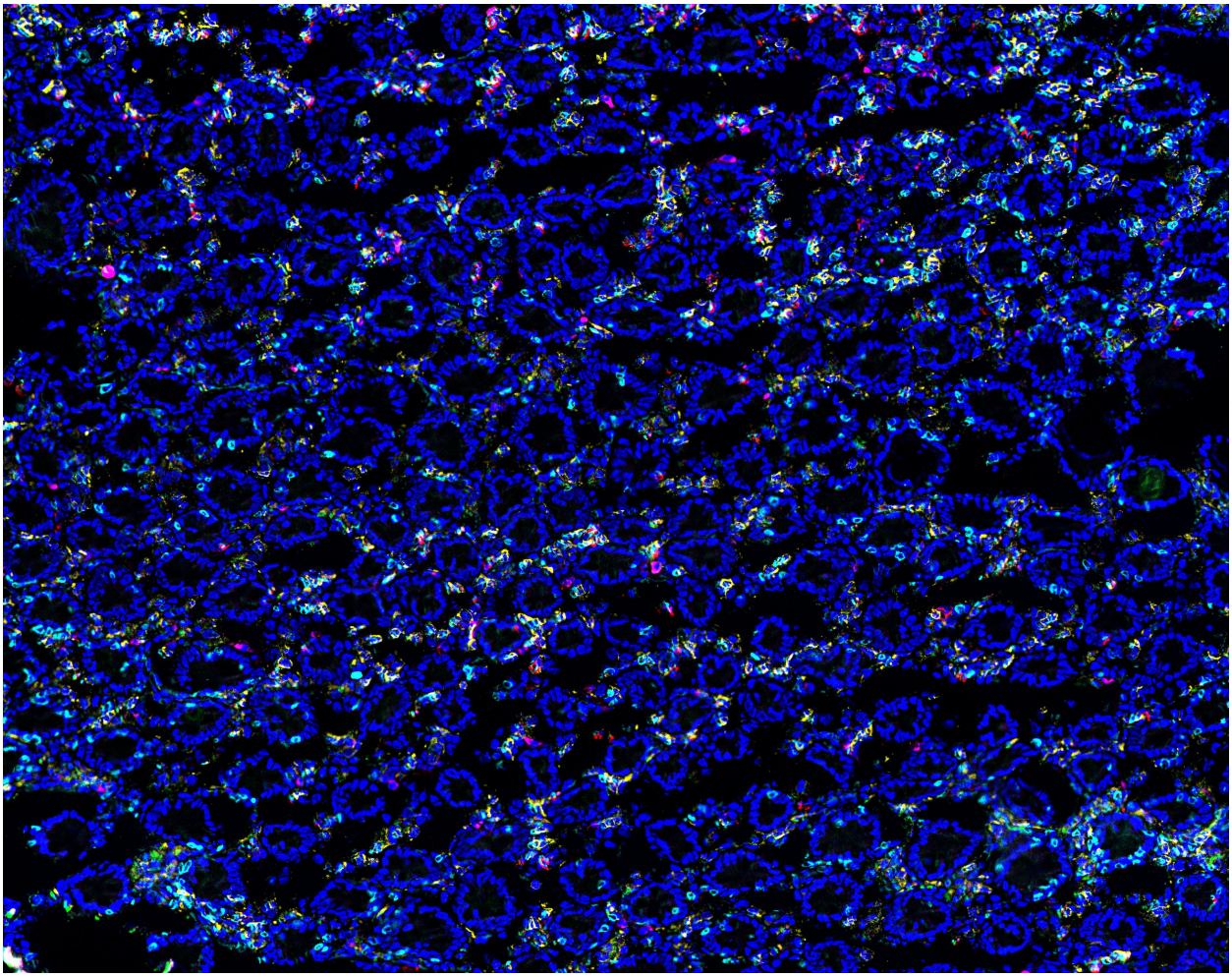
For each segmented cell, `skimage.measure.regionprops_table` extracts morphology (area, centroid, eccentricity, major/minor axis length, solidity) and intensity (mean and max) per channel. The result is a tidy DataFrame, one row per cell, ready for pandas / gating / clustering.

3.5 Save

- `<name>_masks.tif` — int32 instance mask, opens in Fiji / napari / QuPath.
- `<name>_cells.csv` — per-cell feature table.
- `<name>_overlay.png` — three-panel figure (input, mask, outlined overlay).
- `all_cells.csv` — combined across a batch run.

4. Demo image

The example image included in `examples/fluorescence_sample.png` is a densely-packed tissue section with DAPI-stained nuclei (blue) plus several marker channels (cyan, yellow, magenta, red, green) — the kind of input where pre-Cellpose-SAM pipelines would typically require channel selection, diameter estimation, and possibly per-channel denoising. With Cellpose-SAM, the image is passed in as-is.



Demo image: multichannel immunofluorescence. Blue = DAPI / nuclei; cyan, yellow, magenta, red, green = marker channels.

5. Performance expectations

Runtime depends almost entirely on hardware. Paper Table S1, on a 1200×1200 image:

Hardware	Approx. runtime per image
NVIDIA A100	0.4 s
NVIDIA RTX 5090	1 s
NVIDIA RTX 4070 Super	3.2 s
NVIDIA RTX 4060	6 s
Google Colab T4 (free)	7 s
CPU	minutes — not recommended

GPU memory is around 2–3 GB for 1200×1200, scaling roughly with the number of tiles. Cellpose-SAM has internal tiling so very large images don't OOM the GPU — they just take longer.

6. When to look beyond out-of-the-box Cellpose-SAM

Cellpose-SAM is a generalist. It is excellent zero-shot on typical mammalian and plant cell images. It can still benefit from fine-tuning when:

- the imaging modality is very unusual (e.g. novel super-resolution methods),
- cells are extremely large (>120 px) or extremely small (<7.5 px),
- targets are non-cellular (e.g. organelles, bacteria with extreme morphology),
- the data has labeled ground-truth and you want to push average precision higher.

Fine-tuning needs as few as 10–500 ROIs (paper Fig 4) and is documented in the official `train_Cellpose-SAM.ipynb` notebook in the MouseLand/cellpose repo. It runs on consumer GPUs with as little as 8 GB VRAM.

7. Reproducing this work

Three commands, on any reasonably modern machine:

```
git clone https://github.com/<you>/cellpose-sam-automation
cd cellpose-sam-automation && pip install -r requirements.txt
jupyter lab notebooks/cellpose_sam_demo.ipynb
```

8. Citation

Pachitariu M, Rariden M, Stringer C. *Cellpose-SAM: superhuman generalization for cellular segmentation.* bioRxiv 2025.04.28.651001 (2025).

License: code is Apache-2.0. Cellpose-SAM model weights are CC-BY-NC.